SKYTRAC

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Hackathon

February 28 – March 7, 2026

Lindi Sale
HR Manager
lsale@skytrac.ca

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Overview

The Big Picture

1. Welcome – Malachi Nordine, Company President
2. General Information – Lindi Sale, HR Manager
3. Hack-A-Thon Challenge Statement - Gilles Bessens, VP Engineering
4. Hardware Briefing – Andre Lamprecht, Senior Engineering Manager
5. Software Briefing – Steve Lepage, IT & Software Development Manager
6. Evaluation Criteria – Iain Ronis, Director - Product Management

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Welcome from the President

Malachi Nordine

AVIATION'S CONNECTIVITY PLATFORM

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What You Need To Know

Lindi Sale – HR Manager

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General

Launch Day Schedule

Launch Day – February 28, 2026	
9:30 AM – 10:00 AM	Registration & Check-In
10:00AM – 11:15 AM	Welcome & Kickoff (With the President, Engineering Leadership, Product Management and HR)
11:15 AM	Hackathon Begins
12:30 PM – 1:00 PM	Lunch Break
4:00PM	Day 1 Wrap-Up (Official agenda ends, but the office remains open until 6:00pm for those who wish to continue working)

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General

Office Access

Dates	Time
Opening Day - Saturday	9AM - 6PM
Sunday (by prior arrangement)	9AM – 4PM
Week days (by prior arrangement)	9AM - 7PM
Closing Day - Saturday	9AM - 5PM

- To arrange office access during the week, contact:

Lindi Sale, HR Manager, at lsale@skytrac.ca or 778-583-6179, doing so at least 24 hours in advance.

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General

Closing Day Schedule

Closing Day – March 7, 2026	
9:30 AM – 10:00 AM	Check in
10:00AM – 12:00 PM	Hackathon winds down – Finalise presentation
11:30 PM	Working lunch
12:00 PM – 2:30 PM	Team presentations
2:30 – 3:00PM	Judging
3:00 – 4:00PM	Closing ceremony and Social Hour

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General

Key Contacts

Team Name	Leader	Email
UBComms (UBCO)	Steve Lepage	slepage@skytrac.ca
The Forbidden Lan (Carleton)	Mathias Girouard	mgirouard@skytrac.ca
QWERTY (Carleton) + Walkio (U-Ottawa)	Sang Nguyen	samderlust@gmail.com

- Please plan to touch base with your Company leader **at least once each day during office hours**. These quick check-ins are a great opportunity to ask questions, validate direction, and get feedback early.

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General

Housekeeping

- Don’t plug into any docking stations or Ethernet
- Washrooms
- Emergency Exits, Muster Point
- Designated team workspaces
  At your team workspace you have:
  - 1 SKYLINK test unit
  - 1 DigitalOcean Account login
  - Guest Network Wifi Password
  - Your team's key contact & HR contact to arrange access
  - Emergency Exits & Muster Point
  - SKYTRAC Hack-A-Thon Nike Backpack + Tote bag for each participant

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Hack-A-Thon Challenge Statement

Gilles Bessens – VP of Engineering

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Mission

Challenge Statement

The Challenge:

- Aircrew teams operate in remote regions.
- Reliable communication is essential for safety and coordination.
- Existing satellite networks provide coverage but have:
    - High latency
    - Low bandwidth
    - Expensive data usage
- Traditional Push-To-Talk (PTT) systems are ineffective over these satellite constraints.
- Teams lack reliable, low-latency, group-based communication.

Your Mission:

- Build a mobile application that enables walkie-talkie–style Push-To-Talk (PTT) communication over satellite networks.
- Ensure the solution works effectively in remote areas.
- Support group-based communication.
- Require all participants to register and join talkgroups
- Enable voice and text message exchange within talkgroups.

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Mission

Challenge Statement

What You'll Build:

Participants are challenged to design a system that:

- Should feel like talking on a Walkie Talkie radio
- Provides low-latency PTT communication over high-latency satellite networks
- Minimizes data usage using high compression and lightweight protocols
- Chooses the best architecture and framework for the application
- Supports centralized server distribution *or* multicast-style decentralized message fan-out
- Includes a web portal to activate devices, and manage authentication/permissions

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Mission

Challenge Statement

Documentation

Present your preliminary design, including use cases, architecture and draft user interface.

Bonus Challenge

- Seamless group switching via intuitive UI
- Avoid walk-ons when multiple people talk at once
- Background monitoring of multiple channels
- Efficient handling of keys and encryption without excess data usage
- Implement a moving map that shows the location of all the talk group participants
- Talk to an ICOM gateway to talk to other radio systems

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Hardware Briefing

Andre Lamprecht – Senior Manager, Hardware Engineering

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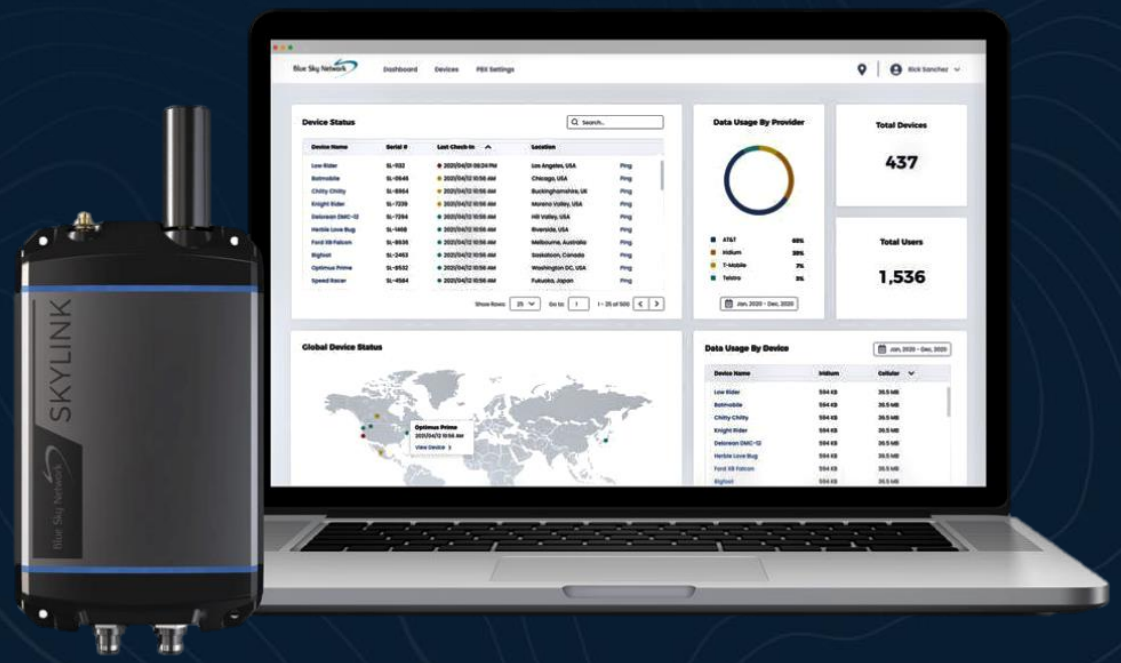
Hardware

SKYLINK Overview

- **Internet Connectivity Module.**
  - **Iridium Certus**
  - **Cellular**
- **Operates over Cellular (LTE) and over Iridium Certus when not in cellular range.**
- **Configured and controlled over Web interface**



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Hardware

Key Specifications

Dimensions	8”L x 5”W x 1.5”H (12.7 x 20.3 x 3.2 cm)
Weight	1.5lbs. (0.7kg)
Power Input	10-34 VDC
Power Consumption	7W Nominal, 18W Peak
Operational Temperature	-40°C - +70°C
Storage Temperature	-40°C - +85°C
Operational Humidity	Less than or equal to 75% RH
Storage Humidity	Less than or equal to 93% RH
Satellite Frequency	1616 MHz – 1626.5 MHz
Cellular Frequency	700 MHz – 2700 MHz
Iridium Certus 100 Throughput	88 Kbps downlink / 22 Kbps uplink
LTE-A Cat 12 Throughput	600 Mbps downlink / 150 Mbps uplink
Wi-Fi Frequency	2.4 GHz / 5 GHz (Configurable)
Wi-Fi Standard	IEEE 802.11 ac/a/b/g/n
Ethernet Interface	CAT6
Serial Interface	RS232 , USB

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Hardware

Interfaces

Power

Cellular SMA Female Connector

USB and RS232 adapters

Iridium Satellite Antenna
SMA Female

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Software Briefing

Steve Lepage - IT & Software Development Manager

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Software Design Pattern

Technical Requirements:

Mobile App Features

- Push-to-talk button with real-time transmit/receive indicators
- Group registration & membership control
- Performance oriented voice messages
- Ability to function under constrained satellite connections
- Optional peer-to-peer or multicast-like communication to reduce delay

Backend Server or Distributed Architecture

- Central server for message relay or peer-to-peer distribution
- Must preserve message integrity and ensure only authorized users participate
- Support for user/device authentication
- Distributed key management to generate, rotate, and share group keys securely

Web Portal

- Create & configure talkgroups
- Activate, authorize, and assign devices/users
- Override or disable rogue units
- View system status and logs

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Software

Certus Data Flow

- Network Address Translation (NAT)
- Each Certus modem has a WAN internal IP (Static)
- DC has 1 Public IP
- This is the same as your cellphone / cell networks
- Only provides access to the internet no reverse NAT

Iridium Ground Station
Tempe AZ

SKYTRAC
Toronto DC

Target Destination

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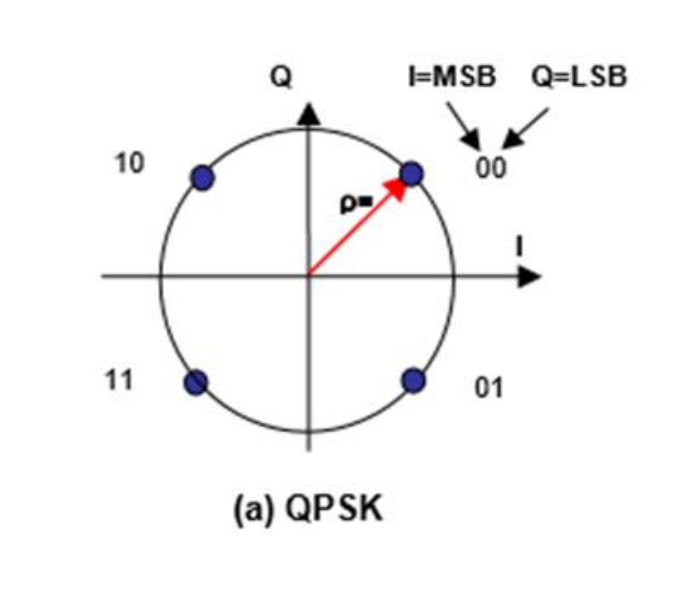
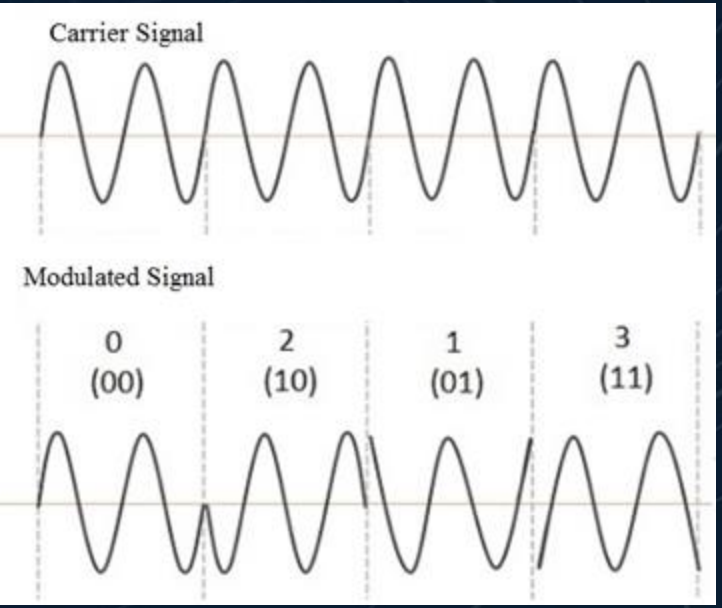
Software

Certus 9770 SFX Modem

- SFX (9770)
  - 2 Voice lines
  - 22Kbps up / 88Kbps down
  - Passive ant
  - 20+W Output
  - 1 Channel / 1 TX slot (QPSK)
  - Phase Shift





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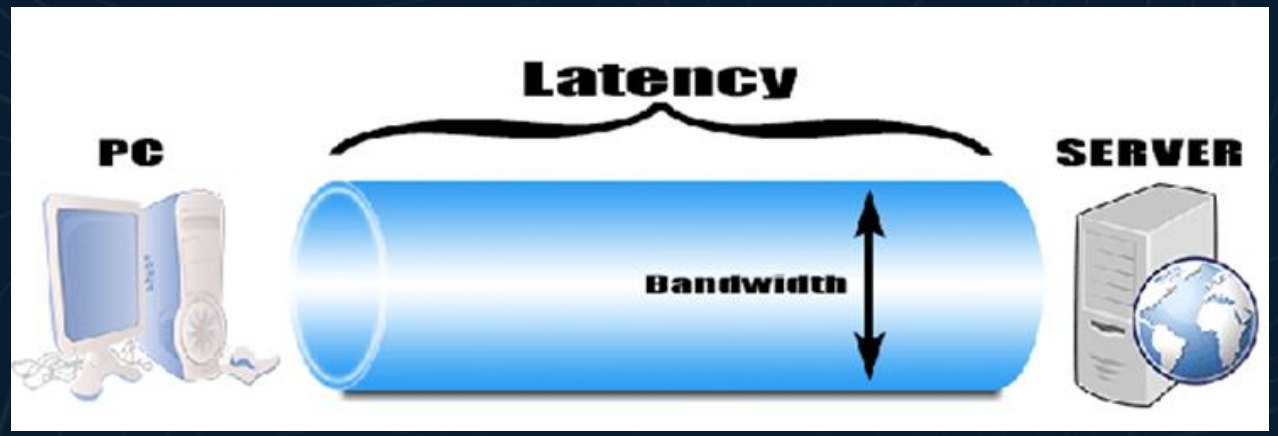
Software

Certus Bandwidth



- Certus Bandwidth is limited (22Kbps / 88Kbps)! Most cell phones are at minimum 20x faster (20Mbps+)
- Think dial up speeds, the birth of the internet, (yes I'm that old)
- With limited bandwidth we need to make good use of the link.
- Many modern devices and PC's will in the background try to connect and download things when they first see internet connectivity

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Troubleshooting

- Certus data connection starts and stops when signal is good or nonexistent.
- Good signal, clear view of the sky needed, partial view of the sky can cause intermittent data (new sat every 8 minutes)
- Initial boot up requires a few minutes warm up of good signal to open the data channel
- Have you tried turning it off and back on again?

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Judging Criteria

Iain Ronis – Director, Product Development

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Final project deliverables

Deliverables on the Final Day:

- Working prototype (mobile app + backend or distributed logic + web portal)
- Requirements, Designs, Use Cases, Architecture diagram(s) and explanation of design tradeoffs, including why the messaging framework was chosen
- Short demo showing talkgroup communication over simulated or real satellite conditions

Teams will have 30 minutes to present their solutions to the Judges on the Closing Day.

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JUDGING CRITERIA

Product Design Hackathon

40% Technical

30% UX

15% Architecture

15% Presentation

+15% Innovation

Technical Functionality

40%

of total score

Push-to-Talk Reliability

Reliable PTT communication under satellite comms constraints

Auth & Access Control

Correct registration, authentication, and talkgroup access control

Data Efficiency

Low data usage through compression and efficient protocols

Message Relay

Successful relay (centralized or decentralized) with minimal delay

User Experience

30% of total score

30%

Intuitive PTT Controls

Simple, intuitive UI for push-to-talk controls

Audio/Visual Feedback

Clear visual and audio feedback for transmitting and receiving

Talkgroup Switching

Smooth switching between multiple talkgroups

Web Portal Usability

Web portal usability and clarity for management tasks

Architecture & Scalability

Architectural Justification

Clear justification of architectural choice (server-based, multicast, hybrid)

Scalability

Scalability across many users and multiple talk groups

Connectivity Handling

Thoughtful handling of satellite bandwidth, latency, and intermittent connectivity

15%

of total score

Presentation & Documentation

15% of total score

15%

1

Problem Articulation

Clear articulation of the problem being solved and its constraints

2

Design & Diagrams

Well-explained design, diagrams, and trade-offs in your documentation

3

Prototype Demo

Effective demonstration of the working prototype

INNOVATION & BONUS

+15% BONUS

Multi-Group Monitoring

Multi-group monitoring & switching

Advanced Compression

Advanced compression, custom protocol, or novel routing techniques

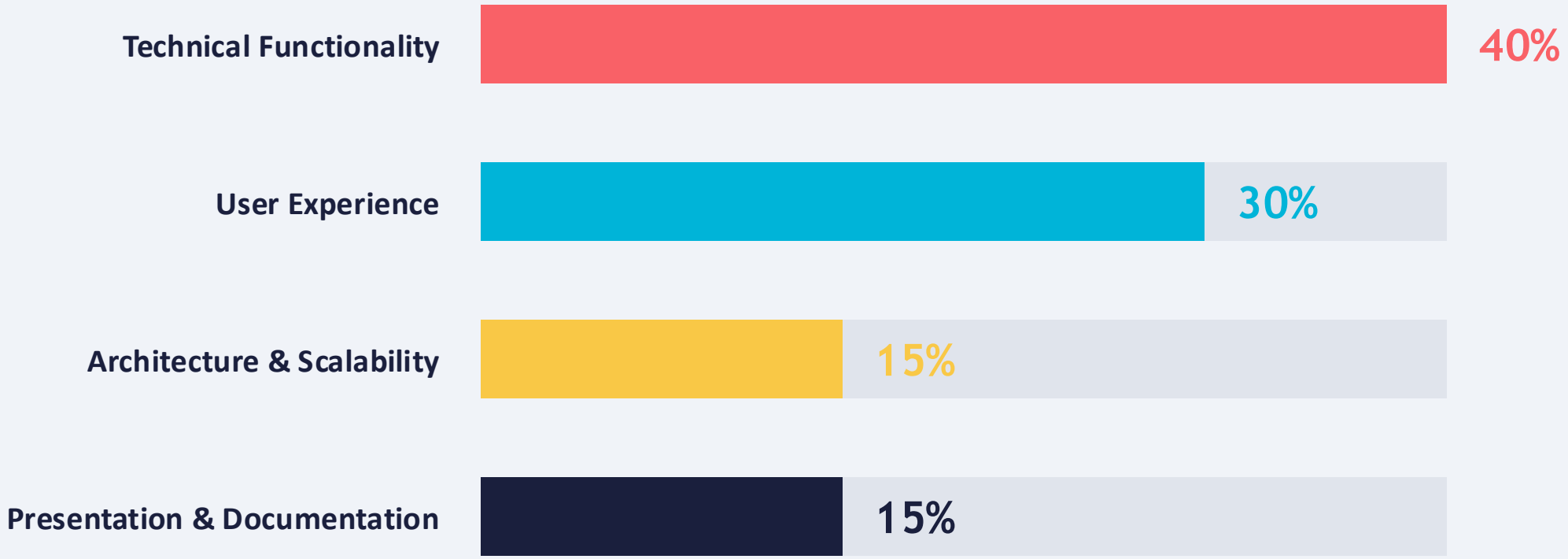
Creative Optimization

Creative approaches to reducing latency or bandwidth usage

ICOM Gateway

ICOM gateway testing and integration

Score Breakdown



Innovation & Bonus Features

+15%

Total: 100% + 15% Bonus

2025 Winners: CU Telecom - Carleton

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We are so pleased you have decided to join the Hackathon and build the future of remote communications!

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